Ackermann-like car model Autonomous modules

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***Abstract*—Autonomous vehicles rely on a combination of advanced control, localization, and navigation modules to ensure safe and efficient operation. This paper presents a literature review on recent advancements (2020-present) in the autonomy modules of Ackermann-like car models. The review focuses on four key aspects: (1) Control modules for speed regulation and lateral movement, including cruise control for lane keeping and lateral control for lane changing; (2) Localization techniques emphasizing sensor filtration and state estimation; (3) Navigation and planning strategies for path optimization and obstacle avoidance; and (4) Integrated autonomous car architectures that connect all modules. The findings provide insight into the latest research trends and technological advancements in the field, highlighting challenges and potential future directions for fully autonomous vehicle development.**

***Index Terms*—Autonomous Vehicles, Localization, Sensor Fusion, Vehicle-to-Everything (V2X), SLAM, Real-Time Positioning**

I. Introduction

Autonomous vehicles (AVs) rely on localization, path planning, and control for safe navigation. Localization determines the vehicle's precise position using sensor-based methods (LiDAR, GPS, IMU, vision), V2X communication, and sensor fusion. Challenges like sensor noise and GPS loss require robust fusion algorithms. Path planning generates optimal trajectories considering road geometry, traffic, and obstacles, leveraging sampling- and optimization-based approaches. Finally, the control module ensures the vehicle follows the planned path via PID, MPC, and AI-driven controllers. The interplay of these modules is crucial for robust autonomy, and this paper explores their latest advancements and challenges.

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II. Literature Review on Autonomous Vehicle Control, Localization, and Path Planning

A. Control Module

Autonomous vehicle control modules are responsible for maintaining stability, safety, and efficiency during operation. These modules include lane-keeping systems (LKS) and lane-changing maneuvers, both of which rely on sensor fusion, AI-driven decision-making, and advanced control algorithms.

1) Lane-Keeping Systems (LKS): LKS ensures the vehicle remains within its designated lane using vision-based sensors, GPS data, and IMUs. Traditional methods employ geometric algorithms for steering corrections, while modern approaches incorporate deep learning-based perception for enhanced robustness. Model Predictive Control (MPC) optimizes steering inputs by predicting future vehicle states, whereas bio-inspired techniques such as fuzzy logic and genetic algorithms improve adaptability to complex environments.

2) Lane-Changing Maneuvers: Lane-changing is a dynamic control task requiring real-time trajectory planning and obstacle detection. Methods such as polynomial fitting, Be´zier curves, and optimal control ensure smooth and collision-free transitions. Safety measures, including time-to-collision (TTC) estimation, adjust vehicle speed and trajectory based on surrounding traffic behavior. Multi-agent reinforcement learning and game theory models improve interaction with human-driven vehicles, while V2X communication facilitates cooperative lane-changing strategies.

3) Key Technologies in Lateral Control: Modern AVs employ sensor fusion techniques combining LiDAR, cameras, radar, and ultrasonic sensors for precise perception. Bayesian inference and Kalman filtering enhance data integration, improving lane detection and obstacle avoidance. Edge computing minimizes latency by processing sensor data locally, while AI-based perception models utilize Transformer architectures for real-time decision-making. V2X communication enhances

safety through decentralized cooperative perception, and cybersecurity measures, such as blockchain protocols, safeguard against cyber threats. Future advancements in neuromorphic computing and bio-inspired control models aim to further enhance lateral control robustness.

B. Localization Module

Accurate localization is essential for autonomous navigation, relying on active and passive sensor-based techniques as well as V2X communication.

1) Active Sensor-Based Localization: Active sensors, such as LiDAR, radar, and ultrasonic sensors, measure distances and detect objects. LiDAR constructs detailed maps but is costly and weather-sensitive. Radar performs well in adverse conditions but has lower spatial resolution. Ultrasonic sensors aid in short-range applications like parking. Hybrid sensor architectures combining LiDAR and radar enhance robustness, while deep learning models refine object detection. Advances in LiDAR (solid-state, MEMS) and radar (MIMO systems) improve localization accuracy and durability. Multi-sensor fusion enhances reliability in high-speed scenarios.

2) Passive Sensor-Based Localization: Passive sensors rely on external signals for positioning. GPS provides global coverage but suffers in urban canyons. IMUs offer high-frequency updates but accumulate drift. Vision-based localization extracts environmental features but struggles in low-light conditions. Event-based cameras reduce computational load, and GNSS augmentation (RTK GPS) improves accuracy. Fusing visual odometry with IMU data enhances dead reckoning performance, ensuring robust localization with minimal infrastructure dependence.

3) V2X-Based Localization: V2X localization enhances positioning through inter-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication. V2V techniques like TOA and AOA improve tracking, while cooperative positioning reduces sensor dependency. Ultra-wideband (UWB) and 6G technologies refine accuracy. V2I methods leverage RSUs, RFID, and UWB signals for enhanced localization. Edge computing optimizes data processing, and AI-driven modeling compensates for signal losses, improving real-time positioning.

4) Data Fusion for Localization: Multi-sensor fusion integrates GPS, LiDAR, radar, and IMU data using algorithms like EKF and Particle Filters. AI-driven fusion dynamically refines estimates, leveraging deep learning for enhanced accuracy. High-definition maps aid localization, with SLAM-based approaches enabling real-time mapping. Crowdsourced self-updating maps and semantic mapping improve adaptability, while lightweight SLAM algorithms optimize real-time performance on embedded AV systems.

C. Path Planning Module

Path planning enables AVs to compute safe and efficient trajectories while avoiding obstacles. Approaches range from classical algorithms to heuristic and learning-based methods.

1) Classical Approaches: Graph-based methods, such as Dijkstra's and A*, efficiently find paths in structured environments but struggle with scalability and dynamic obstacles. Sampling-based methods, including RRT and PRM, generate paths in complex settings but may not always be optimal. Enhanced versions like RRT* refine path quality, though classical methods remain computationally intensive for real-time adaptation.

2) Heuristic Approaches: Heuristic methods optimize paths through bio-inspired techniques. Genetic Algorithms (GA) evolve solutions via selection and mutation, balancing multiple objectives. Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO) simulate natural behaviors to iteratively refine paths. These techniques handle multi-objective optimization well but require fine-tuned parameters and significant computational power.

3) Learning-Based Approaches: Machine learning enhances path planning by adapting to dynamic environments. Supervised learning trains models for obstacle avoidance but requires extensive data. Reinforcement Learning (RL), including Deep Q-Networks (DQN) and Actor-Critic methods, enables AVs to learn policies through experience. Deep Reinforcement Learning (DRL) integrates neural networks for high-dimensional navigation, though high computational demands limit real-time use.

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